

▼ Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv('AirPassengers.csv')
```

```
df.head()
```



	Month	#Passengers	
0	1949-01	112	
1	1949-02	118	
2	1949-03	132	
3	1949-04	129	
4	1949-05	121	

Next steps:

[Generate code with df](#)
[View recommended plots](#)
[New interactive sheet](#)

▼ EDA

```
df.info()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   Month      144 non-null    object
 1   #Passengers 144 non-null    int64
dtypes: int64(1), object(1)
memory usage: 2.4+ KB
```

```
df.shape
```



```
(144, 2)
```

```
df['Month']=pd.to_datetime(df['Month'])
```

```
df.info()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   Month      144 non-null    datetime64[ns]
 1   #Passengers 144 non-null    int64
dtypes: datetime64[ns](1), int64(1)
memory usage: 2.4 KB
```

```
df.isnull().sum().sum()
```



```
0
```

```
df.duplicated().sum()
```



```
0
```

```
df.index=df['Month']
```

```
df.drop('Month',axis=1,inplace=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 144 entries, 1949-01-01 to 1960-12-01
Data columns (total 1 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   #Passengers  144 non-null    int64
dtypes: int64(1)
memory usage: 2.2 KB
```

```
df
```

Month	#Passengers
1949-01-01	112
1949-02-01	118
1949-03-01	132
1949-04-01	129
1949-05-01	121
...	...
1960-08-01	606
1960-09-01	508
1960-10-01	461
1960-11-01	390
1960-12-01	432

144 rows × 1 columns

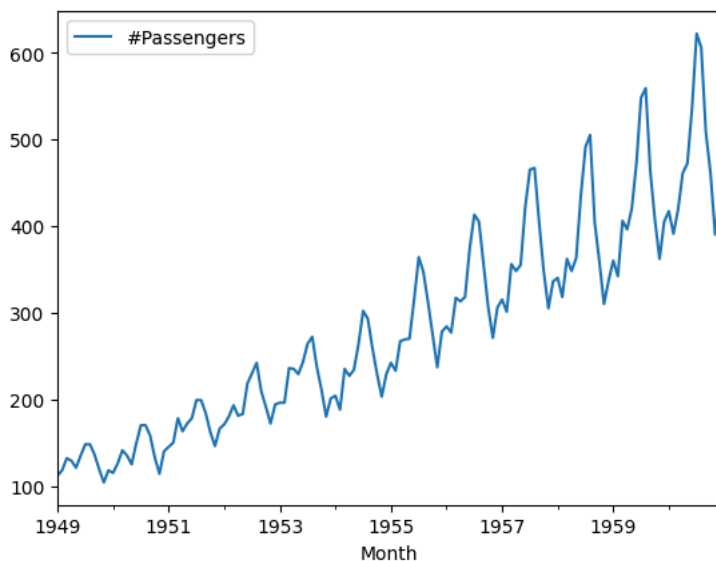
Next steps:

[Generate code with df](#)
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[New interactive sheet](#)

✓ Vizualizing the data

```
df.plot()
```

```
<Axes: xlabel='Month'>
```



✓ Adfuller Test

```
#adf --> Adfuller test
#H0 --> data is stationary
#H1 --> data is not stationary
```

```
from statsmodels.tsa.stattools import adfuller
```

```
result=adfuller(df['#Passengers'])
```

```
result
```

```
(0.8153688792060498,
 0.991880243437641,
 13,
 130,
 {'1%': -3.4816817173418295,
  '5%': -2.8840418343195267,
  '10%': -2.578770059171598},
 996.692930839019)
```

```
p_val=result[1]
p_val
```

```
0.991880243437641
```

```
if(p_val<0.05):
    print("Accept the null hypothesis,The time Series data is stationary")
else:
    print("Accept the alternate hynothesis,The Time series data is not stationary")
```

```
Accept the alternate hynothesis,The Time series data is not stationary
```

▼ Differencing

```
rolling_mean=df.rolling(window=12).mean()
```

```
rolling_mean.head(12)
```

```
#
```

Month	#Passengers
1949-01-01	NaN
1949-02-01	NaN
1949-03-01	NaN
1949-04-01	NaN
1949-05-01	NaN
1949-06-01	NaN
1949-07-01	NaN
1949-08-01	NaN
1949-09-01	NaN
1949-10-01	NaN
1949-11-01	NaN
1949-12-01	126.666667

Next steps:

[Generate code with rolling_mean](#)
[View recommended plots](#)
[New interactive sheet](#)

```
df_detrended=df-rolling_mean
```

```
df_detrended
```

#Passengers		
Month		
1949-01-01	NaN	
1949-02-01	NaN	
1949-03-01	NaN	
1949-04-01	NaN	
1949-05-01	NaN	
...	...	
1960-08-01	142.666667	
1960-09-01	40.916667	
1960-10-01	-10.583333	
1960-11-01	-83.916667	
1960-12-01	-44.166667	

144 rows × 1 columns

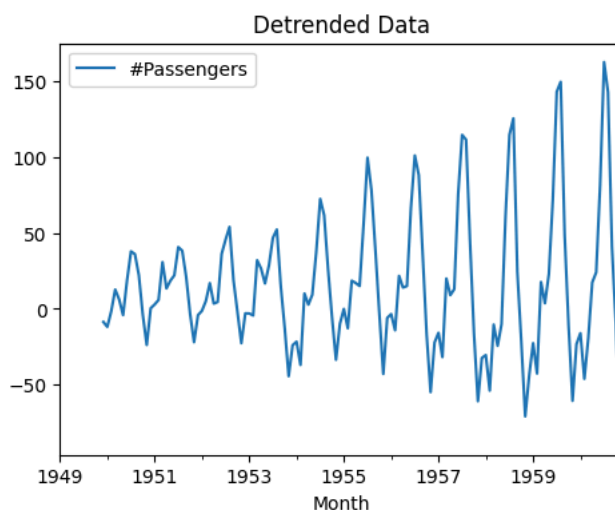
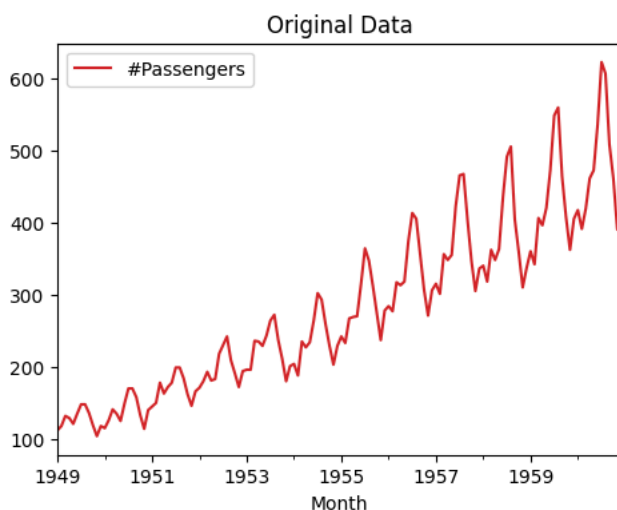
Next steps:

[Generate code with df_detrended](#)[View recommended plots](#)[New interactive sheet](#)

```
ax1=plt.subplot(1,2,1)
df.plot(figsize=(12,4),color='tab:red',title='Original Data',ax=ax1)

ax2=plt.subplot(1,2,2)
df_detrended.plot(figsize=(12,4),color='tab:blue',title='Detrended Data',ax=ax2)
```

```
<Axes: title={'center': 'Detrended Data'}, xlabel='Month'>
```



```
df_detrended.dropna(inplace=True)
```

```
result=adfuller(df_detrended['#Passengers'])
```

result

```
(-3.164968129955144,
 0.022104139473878875,
 13,
 119,
 {'1%': -3.4865346059036564,
  '5%': -2.8861509858476264,
  '10%': -2.579896092790057},
 906.2878851500943)
```

```
p_val=result[1]
```

```

if p_val<0.05:
    print("Accept the null hypothesis,The time Series data is stationary")
else:
    print("Accept the alternate hypothesis,The Time series data is not stationary")

```

↗ Accept the null hypothesis,The time Series data is stationary

✓ Plotting the stationary data to check

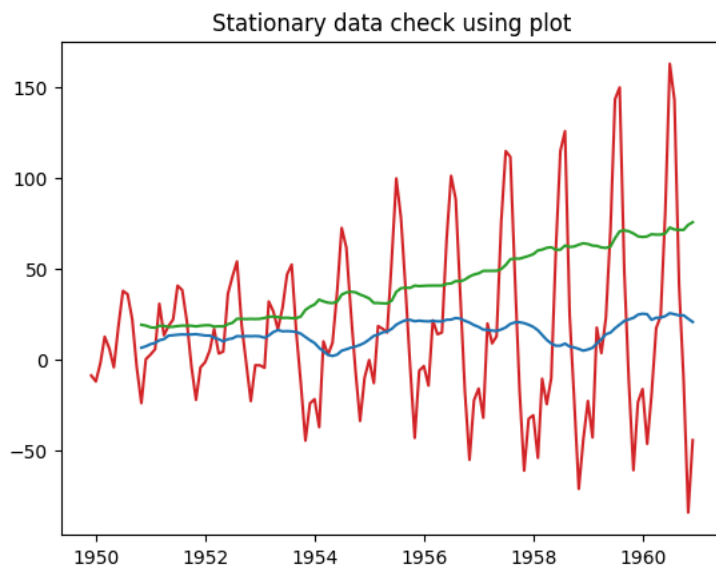
```

m=df_detrended.rolling(window=12).mean()
s=df_detrended.rolling(window=12).std()

plt.plot(df_detrended,color='tab:red',label='Original')
plt.plot(m,color='tab:blue',label='Rolling Mean')
plt.plot(s,color='tab:green',label='Rolling Std')
plt.title('Stationary data check using plot')
plt.show()

```

↗



✓ Model Building

```

train=df_detrended[:120]['#Passengers']
test=df_detrended[120:]['#Passengers']

```

train



#Passengers	
Month	
1949-12-01	-8.666667
1950-01-01	-11.916667
1950-02-01	-1.583333
1950-03-01	12.666667
1950-04-01	6.166667
...	
1959-07-01	143.083333
1959-08-01	149.583333
1959-09-01	48.666667
1959-10-01	-11.333333
1959-11-01	-60.666667

120 rows × 1 columns

dtype: float64

test



#Passengers	
Month	
1959-12-01	-23.333333
1960-01-01	-16.083333
1960-02-01	-46.166667
1960-03-01	-19.250000
1960-04-01	17.333333
1960-05-01	24.000000
1960-06-01	81.750000
1960-07-01	162.583333
1960-08-01	142.666667
1960-09-01	40.916667
1960-10-01	-10.583333
1960-11-01	-83.916667
1960-12-01	-44.166667

dtype: float64

```
from statsmodels.tsa.arima.model import ARIMA

model=ARIMA(train,order=(1,0,2))
model_fit=model.fit()

df_detrended['Predicted']=model_fit.predict(start=len(train),end=len(train)+len(test)-1)

/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No frequency information was provided, so i
self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No frequency information was provided, so i
self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No frequency information was provided, so i
self._init_dates(dates, freq)

df_detrended
```

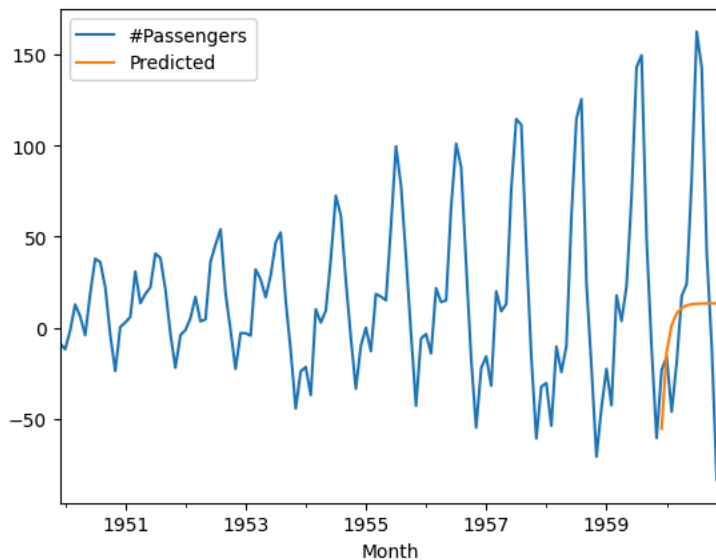
	#Passengers	Predicted
Month		
1949-12-01	-8.666667	NaN
1950-01-01	-11.916667	NaN
1950-02-01	-1.583333	NaN
1950-03-01	12.666667	NaN
1950-04-01	6.166667	NaN
...	...	...
1960-08-01	142.666667	13.223011
1960-09-01	40.916667	13.280637
1960-10-01	-10.583333	13.306771
1960-11-01	-83.916667	13.318622
1960-12-01	-44.166667	13.323997

133 rows × 2 columns

Next steps: [Generate code with df_detrended](#) [View recommended plots](#) [New interactive sheet](#)

```
df_detrended[['#Passengers', 'Predicted']].plot()
```

<Axes: xlabel='Month'>



▼ auto_arima

```
#!pip install numpy==1.26.4 scipy==1.14.1
```

```
Requirement already satisfied: numpy==1.26.4 in /usr/local/lib/python3.12/dist-packages (1.26.4)
Collecting scipy==1.14.1
  Downloading scipy-1.14.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (60 kB)
  Downloading scipy-1.14.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (40.8 MB)
  40.8/40.8 MB 63.3 MB/s 0:00:00
```

```
Installing collected packages: scipy
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.1
    Uninstalling scipy-1.16.1:
      Successfully uninstalled scipy-1.16.1
  Successfully installed scipy-1.14.1
```

WARNING: The following packages were previously imported in this runtime:
[scipy]
You must restart the runtime in order to use newly installed versions.

RESTART SESSION

```
!pip install pmdarima
```

```

Requirement already satisfied: pmdarima in /usr/local/lib/python3.12/dist-packages (2.0.4)
Requirement already satisfied: joblib>=0.11 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (1.5.1)
Requirement already satisfied: Cython!=0.29.18,!=0.29.31,>=0.29 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (3.0.12)
Requirement already satisfied: numpy>=1.21.2 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (1.26.4)
Requirement already satisfied: pandas>=0.19 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (2.2.2)
Requirement already satisfied: scikit-learn>=0.22 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (1.6.1)
Requirement already satisfied: scipy>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (1.14.1)
Requirement already satisfied: statsmodels>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (0.14.5)
Requirement already satisfied: urllib3 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (2.5.0)
Requirement already satisfied: setuptools!=50.0.0,>=38.6.0 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (80.9.0)
Requirement already satisfied: packaging>=17.1 in /usr/local/lib/python3.12/dist-packages (from pmdarima) (25.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.19->pmdarima) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.19->pmdarima) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.19->pmdarima) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=0.19->pmdarima) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22->pmdarima) (3.6.0)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels>=0.13.2->pmdarima) (1.0.1)

```

```
from pmdarima import auto_arima
```

```
auto=auto_arima(df_detrended['#Passengers'])
```

```

/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
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warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
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/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
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/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_f
warnings.warn(

```

```
auto.summary()
```




SARIMAX Results

Dep. Variable: y No. Observations: 133
 Model: SARIMAX(3, 0, 2) Log Likelihood: -612.905
 Date: Wed, 27 Aug 2025 AIC: 1239.810
 Time: 16:14:47 BIC: 1260.043
 Sample: 12-01-1949 HQIC: 1248.032
 - 12-01-1960

Covariance Type: opg

	coef	std err	z	P> z	[0.025	0.975]
intercept	6.8579	1.701	4.033	0.000	3.525	10.191
ar.L1	0.8625	0.171	5.045	0.000	0.527	1.198
ar.L2	0.2071	0.236	0.878	0.380	-0.255	0.670
ar.L3	-0.5239	0.142	-3.683	0.000	-0.803	-0.245
ma.L1	0.0684	0.138	0.496	0.620	-0.202	0.338
ma.L2	-0.7564	0.091	-8.293	0.000	-0.935	-0.578
sigma2	577.4777	71.847	8.038	0.000	436.660	718.295

Ljung-Box (L1) (Q): 0.02 Jarque-Bera (JB): 8.11
 Prob(Q): 0.89 Prob(JB): 0.02
 Heteroskedasticity (H): 6.21 Skew: 0.56
 Prob(H) (two-sided): 0.00 Kurtosis: 3.48

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex step)

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
```

```
model=SARIMAX(train,order=(3,0,2),seasonal_order=(3,0,2,12))
model_fit=model.fit()
```

```
df_detrended['Predicted']=model_fit.predict(start=len(train),end=len(train)+len(test)-1)
```

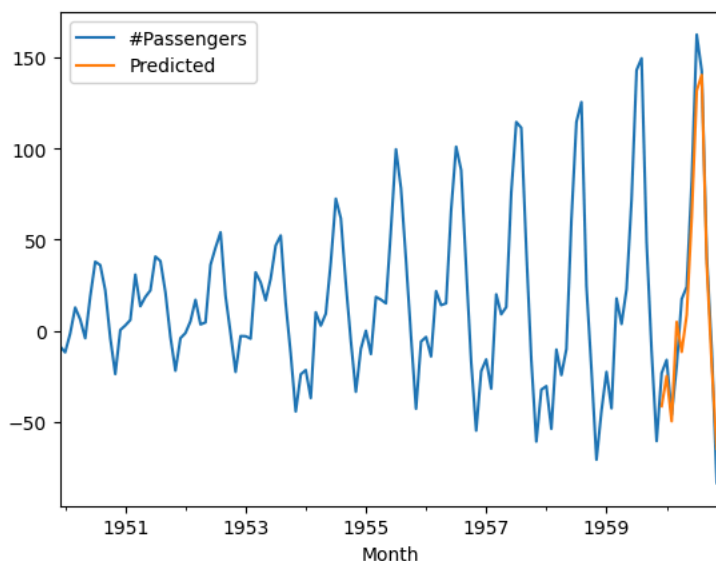


```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No frequency information was provided, so i
self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: No frequency information was provided, so i
self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:966: UserWarning: Non-stationary starting autoregressive p
warn('Non-stationary starting autoregressive parameters')
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA parameters fc
warn('Non-invertible starting MA parameters found.')
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:997: UserWarning: Non-stationary starting seasonal autoreg
warn('Non-stationary starting seasonal autoregressive')
/usr/local/lib/python3.12/dist-packages/statsmodels/base/model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to con
warnings.warn("Maximum Likelihood optimization failed to "
```

```
df_detrended[['#Passengers', 'Predicted']].plot()
```



```
<Axes: xlabel='Month'>
```



```
forecast=model_fit.get_forecast(steps=24) #test+predicting years  
future_preds=forecast.predicted_mean
```

```
df_detrended.plot()  
future_preds.plot()
```

↔ <Axes: xlabel='Month'>

